Custom ESP32 Board with MPU-6050

Your Name

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1 Problem Statement 1: Schematic Design

1.1 Objective

Design the schematic of a custom ESP32 board with an MPU-6050 motion sensor that:

- can be programmed over a USB Type-C port,
- can be powered from either the Type-C port or an external battery,
- integrates the MPU-6050 over I^2C, and
- uses only the components essential for these functions.

1.2 Schematic

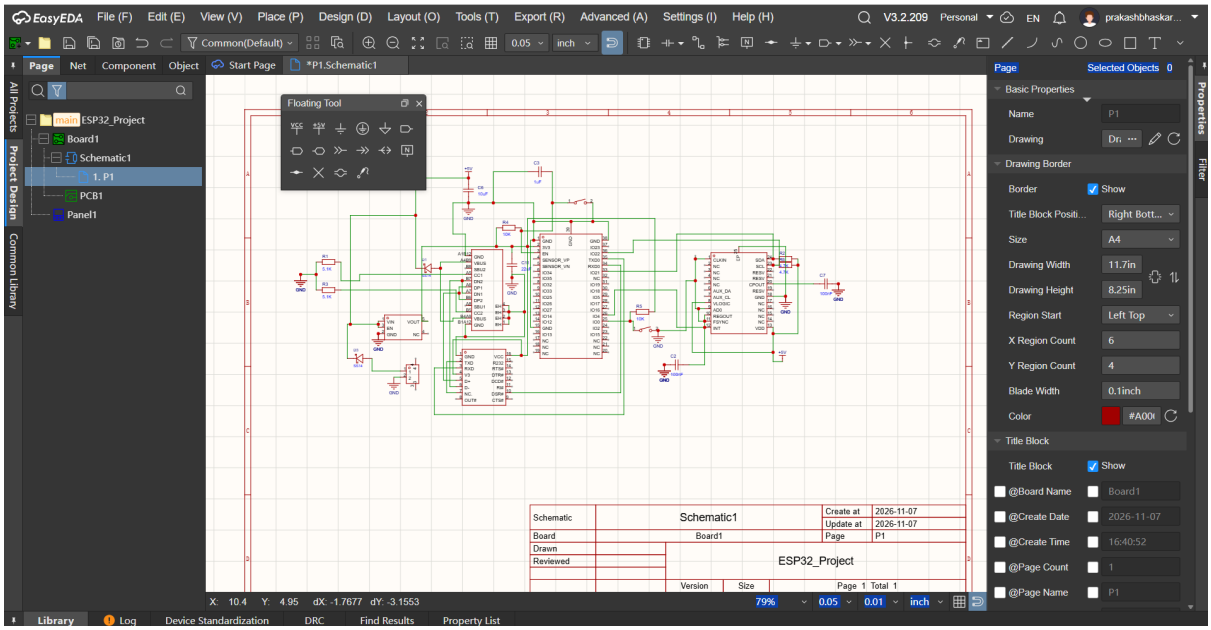


Figure 1: Schematic of the custom ESP32 board with MPU-6050, designed in EasyEDA.

1.3 Main Components

Block	Component	Purpose
Microcontroller	ESP32-WROOM-32E	Main processor with Wi-Fi and Bluetooth
USB connector	USB Type-C 16-pin	5 V power and data
USB-to-serial	CH340C	Code upload over USB
Regulator	AP2112K-3.3	Converts 5 V to 3.3 V
Power selection	2× SS14 Schottky diodes	Selects USB or battery, prevents back-feed
Battery input	2-pin JST connector	External battery
Sensor	MPU-6050	6-axis accelerometer and gyroscope
User controls	2× tactile switches	RESET (EN) and BOOT (IO0)
Passives	Resistors and capacitors	Pull-ups, CC resistors, decoupling

Table 1: Essential components used in the design.

1.4 Circuit Description

Power supply. The 5 V from the Type-C VBUS pin and the battery input each pass through an SS14 Schottky diode and join at a common 5 V rail. The diodes stop one source from feeding current back into the other. The regulator converts this rail to 3.3 V for the ESP32, CH340C and MPU-6050. Input and output capacitors keep the supply stable.

USB and programming. Two 5.1 kΩ resistors on the CC1 and CC2 pins let the Type-C port deliver 5 V. The USB D+ and D− lines go to the CH340C, which converts them to UART. Its TXD pin connects to the ESP32 RXD0 pin, and its RXD pin connects to the ESP32 TXD0 pin, so code can be uploaded over USB.

Boot and reset. The EN and IO0 pins each have a 10 kΩ pull-up to 3.3 V. A tactile switch on each pin pulls it to ground. A 1 μF capacitor on EN gives a clean reset. Holding BOOT while pressing RESET puts the ESP32 into flash mode.

MPU-6050 integration. The MPU-6050 connects to the ESP32 over I^2C, with SDA on GPIO21 and SCL on GPIO22. Two 4.7 kΩ pull-up resistors hold the lines at 3.3 V. The AD0 pin is tied to ground, which sets the I^2C address to 0x68, and the interrupt pin is routed to GPIO4. The required capacitors are placed on the REGOUT, CPOUT and supply pins.

1.5 Design Tool

The schematic was drawn in EasyEDA Pro, and a design rule check was run to confirm there were no connection errors.

2 Problem Statement 2: PCB Design

2.1 Objective

Create a PCB layout for the schematic in Problem Statement 1, keeping the board as small as possible.

2.2 PCB Layout

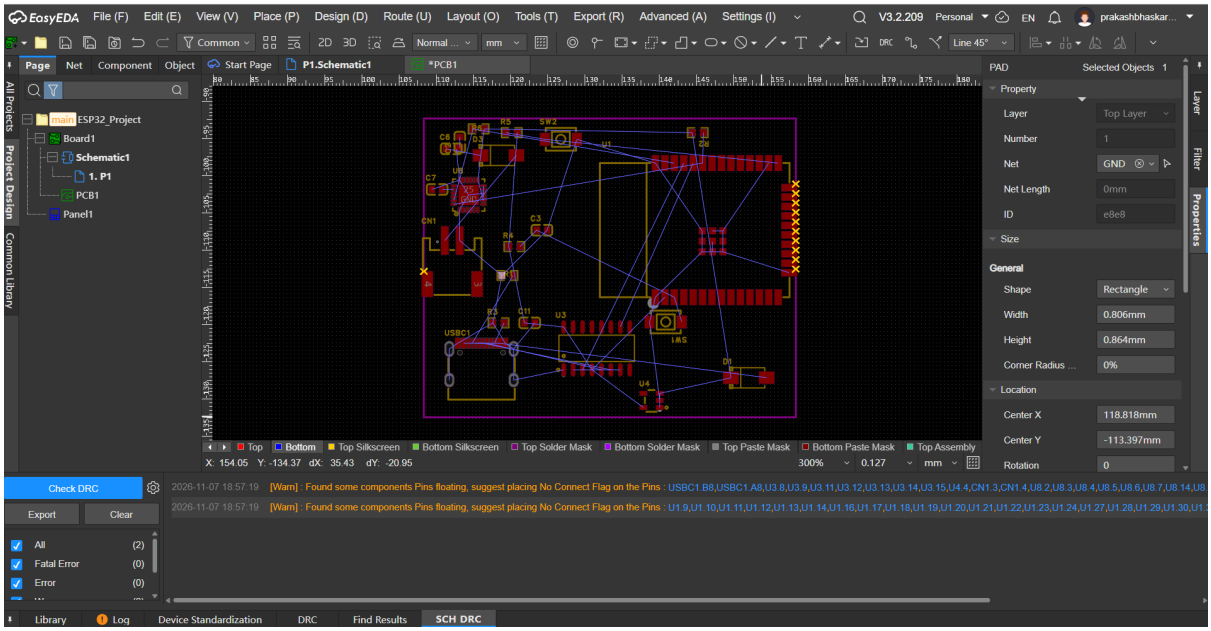


Figure 2: PCB layout of the ESP32 board with MPU-6050 in EasyEDA, with all components placed inside the board outline.

2.3 Design Approach

Board outline. The board outline is drawn on the Board Outline layer as a rectangle of approximately 50 mm × 40 mm, and all components are placed inside it. The board is two-layer, and all components are on the top layer to keep the design simple.

Component placement.

- The **ESP32-WROOM-32E** module is on the right edge, so its antenna end stays at the edge of the board. No other parts are placed near the antenna.
- The **USB Type-C connector** is at the bottom edge, where a cable can be plugged in easily.
- The **CH340C** and the **3.3 V regulator** are close to the USB connector, which keeps the USB data and power tracks short.
- The **MPU-6050** is placed away from the antenna and from the power parts.
- The **RESET and BOOT switches** and the **battery connector** are placed near the board edges for easy access.
- The **decoupling capacitors** are placed next to the pins they supply.

Routing and ground. The connections follow the net list imported from the schematic. Power tracks (5 V and 3.3 V) are made wider than signal tracks, and the USB D+ and D− tracks are kept short and close together. A ground copper pour on the bottom layer gives a low-impedance return path, and no copper is placed under the ESP32 antenna.

Design rule check. The design rule check reported zero errors and zero fatal errors. The only warnings are for intentionally unused pins on the ESP32 and CH340C.

2.4 Conclusion

The schematic and PCB layout meet the stated objectives: code upload over USB Type-C, power from either USB or an external battery, MPU-6050 integration over I^2C, and a minimal set of essential components.